

RV College of Engineering® , Bengaluru – 59
Department of Information Science and Engineering
Database Management Systems (CD252IA)

Mid-Term Report on Database Project

EXPERIENTIAL LEARNING TEAM FINDER		
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1. Tables created details

1. Clusters Table

Purpose: Stores academic groups (e.g., Computing, Engineering).

Key Fields: clusterId (PK), clusterName, branches.

2. Users Table

Purpose: Stores student profiles and academic backgrounds.

Key Fields: userId (PK), name, email, clusterId (FK), branch, year, skills.

Dependency: References Clusters(clusterId).

3. Teams Table

Purpose: Defines the project group and its current status (open/full).

Key Fields: teamId (PK), teamName, projectTitle, createdBy (FK), maxMembers, status.

Dependency: References Users(userId).

4. TeamMembers Table

Purpose: Tracks which users belong to which teams and their specific roles (e.g., Lead, Developer).

Key Fields: memberId (PK), teamId (FK), userId (FK), role, joinedAt.

Dependency: References Teams(teamId) and Users(userId).

5. JoinRequests Table

Purpose: Manages pending applications from users to join specific teams.

Key Fields: requestId (PK), teamId (FK), userId (FK), status (pending/accepted/rejected).

Dependency: References Teams(teamId) and Users(userId).

2. Table creation Queries

1. USERS Table:

```
CREATE TABLE Users (  
    user_id VARCHAR(50) PRIMARY KEY,  
    name VARCHAR(100) NOT NULL,  
    email VARCHAR(100) UNIQUE NOT NULL,  
    cluster VARCHAR(50) NOT NULL,  
    branch VARCHAR(50) NOT NULL  
);
```

2. TEAMS Table

```
CREATE TABLE Teams (  
    team_id VARCHAR(50) PRIMARY KEY,  
    team_name VARCHAR(100) NOT NULL,  
    project_title VARCHAR(150) NOT NULL,  
    project_description TEXT,  
    created_by VARCHAR(50),  
    status VARCHAR(20) DEFAULT 'open',  
    max_members INT DEFAULT 5,  
    FOREIGN KEY (created_by) REFERENCES Users(user_id)  
);
```

3. TEAMMEMBERS Table

```
CREATE TABLE TeamMembers (  
    team_id VARCHAR(50),  
    user_id VARCHAR(50),  
    cluster VARCHAR(50) NOT NULL,  
    PRIMARY KEY (team_id, user_id),  
    FOREIGN KEY (team_id) REFERENCES Teams(team_id),  
    FOREIGN KEY (user_id) REFERENCES Users(user_id)  
);
```

4. JOINREQUESTS Table

```
CREATE TABLE JoinRequests (  
    request_id INT AUTO_INCREMENT PRIMARY KEY,  
    team_id VARCHAR(50),  
    user_id VARCHAR(50),  
    status VARCHAR(20) DEFAULT 'pending',  
    FOREIGN KEY (team_id) REFERENCES Teams(team_id),  
    FOREIGN KEY (user_id) REFERENCES Users(user_id)  
);
```

5. CLUSTERS Table

```
CREATE TABLE Clusters (  
    cluster_id INT AUTO_INCREMENT PRIMARY KEY,  
    cluster_name VARCHAR(50) UNIQUE NOT NULL,  
    description TEXT  
);
```

3. Table Population Queries

1. Users Table :

```
INSERT INTO Users (user_id, name, email, cluster,
branch)
VALUES
('U001', 'Sree Harsha', 'sreeharsha@gmail.com',
'Computer Science', 'CSE'),
('U002', 'Rahul Verma', 'rahul@gmail.com', 'Mechanical',
'Mechanical Engg'),
('U003', 'Ananya Rao', 'ananya@gmail.com',
'Electronics', 'ECE');
```

2. Teams Table :

```
INSERT INTO Teams (team_id, team_name, project_title,
project_description, created_by, status, max_members)
VALUES
('T001', 'Code_Bytes', 'EL Team Finder', 'Platform for
EL project team formation', 'U001', 'open', 5),
('T002', 'MechTronics', 'Smart Automation', 'Automation
using sensors and controllers', 'U002', 'open', 5);
```

3. TeamMembers Table :

```
INSERT INTO TeamMembers (team_id, user_id, cluster)
VALUES
('T001', 'U001', 'Computer Science'),
('T001', 'U002', 'Mechanical'),
('T001', 'U003', 'Electronics');
```

4. JoinRequests Table :

```
INSERT INTO JoinRequests (team_id, user_id, status)
VALUES
('T002', 'U003', 'pending');
```

4. Query execution for data retrieval required for your project

1. View Available Teams in a Specific Cluster

This query helps a student find teams that are still looking for members within their own academic department.

```
SELECT teamName, projectTitle, projectDescription, status
FROM Teams
JOIN Users ON Teams.createdBy = Users.userId
WHERE Users.clusterId = (SELECT clusterId FROM Users WHERE
userId = [Current_User_ID])
AND Teams.status = 'open';
```

2. List All Members of a Specific Team

This is required for the "Team Profile" page to show who is currently in a group and what their roles are.

```
SELECT Users.name, TeamMembers.role, Users.skills,
TeamMembers.joinedAt
FROM TeamMembers
JOIN Users ON TeamMembers.userId = Users.userId
WHERE TeamMembers.teamId = [Target_Team_ID];
```

3. Fetch Pending Join Requests for a Team Lead

A team creator needs to see who has applied to join their team to either accept or reject them.

```
SELECT JoinRequests.requestId, Users.name, Users.branch,  
Users.skills, JoinRequests.requestedAt  
FROM JoinRequests  
JOIN Users ON JoinRequests.userId = Users.userId  
WHERE JoinRequests.teamId = [Lead_Team_ID]  
AND JoinRequests.status = 'pending';
```

4. Get User's Current Team Memberships

This populates the user's dashboard to show which projects they are currently a part of.

```
SELECT Teams.teamName, Teams.projectTitle,  
TeamMembers.role, Teams.status  
FROM Teams  
JOIN TeamMembers ON Teams.teamId = TeamMembers.teamId  
WHERE TeamMembers.userId = [Current_User_ID];
```

5. Search for Students with Specific Skills

Used for the "Find Teammates" feature where a creator looks for students who possess a specific technical skill (e.g., "Python" or "React").

```
SELECT name, email, branch, year, skills  
FROM Users  
WHERE skills LIKE '%React%'  
AND userId != [Current_User_ID];
```